Title: Implement simple Dynamic routing in Cisco Packet Tracer.

AIM: Using a Network Simulator (e.g. packet tracer) configure Dynamic Routing

OBJECTIVES:
To learn various routing protocols and get familiar with network simulator like Cisco Packet Tracer.

SOFTWARES USED:
Cisco Packet Tracer

THEORY:

DYNAMIC ROUTING

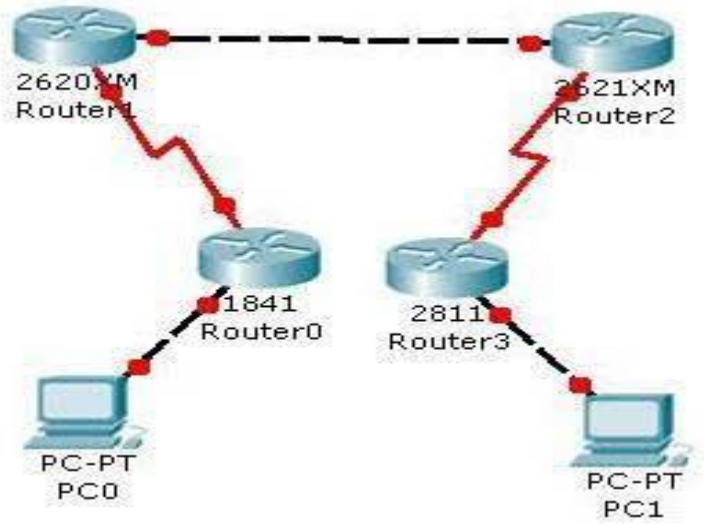
II.RIPv2 Routing Protocol

RIPv2 has one great improvement over RIPv1, instead of sending just the network address in its routing updates it can sends the subnet mask and the next hop address as well, this means that it can advertise non-classful subnets. In this way, RIPv2 supports VLSM and CIDR. Mostly RIPv2 is very similar to RIPv1, it is a Distance Vector routing protocol, it uses hop count as its metric, with a maximum distance of 15 hops.

RIPv2 Differences from RIPv1
- VLSM and CIDR - is supported by sending the subnet mask and the next hop address in its routing updates.
- Multicasts - its routing updates will not be broadcasted to 255.255.255.255 like RIPv1
- Authentication - RIPv2 supports md5 authentication
- Updates - RIPv2 sends and receives version 2 updates only. RIPv1 sends version 1 updates and receives both 1 and 2, however version 2 information is ignored.

RIPv2 Similarities to RIPv1
- Auto Summarizes by default (You will need to turn this off if you have discontinuous networks)
- Distance Vector Protocol
- Hop Count is the metric with a maximum of 15 hops, 16 is infinity and is dropped.

CONFIGURATION:



1841 Series Router0 (R1)		
	FastEthernet0/0	Serial0/0/0
IP address	10.0.0.1	20.0.0.1
Connected With	Pc0	R2 on Serial 0/0
2811 Series Router0 (R4)		
	FastEthernet0/0	Serial0/0/0
IP address	50.0.0.1	40.0.0.2
Connected With	Pc1	R3 on Serial 0/0

2621XM Series Router0 (R3)		
	FastEthernet0/0	Serial0/0/0
IP address	30.0.0.2	40.0.0.1
Connected With	FastEthernet0/0	R4 on Serial 0/0/0
2620XM Series Router1 (R2)		
	FastEthernet0/0	Serial0/0
IP address	30.0.0.1	20.0.0.2
Connected With	R3 on FastEthernet0/0	R1 on Serial 0/0/0

PC-PT PC0		
	FastEthernet0	Default Gateway
IP address	10.0.0.2	10.0.0.1

Connected With	R1 on FastEthernet0/0	
PC-PT PC1		
	FastEthernet0	Default Gateway
IP address	50.0.0.2	50.0.0.1
Connected With	R4 on FastEthernet0/0	

Task 1: To configure and enable rip routing on R1 follow these commands

```
Router>enable
Router#configu
re terminal
Enter configuration commands, one per line.
                                                E
nd with CNTL/Z. Router(config)#hostname R1
R1(config)#interface fastethernet 0/0
R1(config-if)#ip address 10.0.0.1
255.0.0.0 R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
R1(config-if)#exit
R1(config)#interfa
ce serial 0/0/0
R1(config-if)#ip address 20.0.0.1 255.0.0.0
R1(config-if)#clock rate 64000
R1(config-if)#bandwidth 64
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-
router)#network
20.0.0.0 R1(config-
router)#exit
R1(config)#
```

Task2: To configure and enable rip routing on R2 follow these command

```
Router>ena
ble
Router#con
figure
terminal
Enter configuration commands, one per line.
                                                E
```

```
nd with CNTL/Z. Router(config)#hostname R2
R2(config)#interface serial 0/0
R2(config-if)#ip address 20.0.0.2
255.0.0.0 R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed
state to up R2(config-if)#exit
R2(config)#interface fastethernet 0/0 R2(config-if)#ip address
30.0.0.1 255.0.0.0R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to
up
R2(config)#router rip
R2(config-router)#network 20.0.0.0
R2(config-
router)#network
30.0.0.0 R2(config-
router)#exit
R2(config)#
```

Task3: To configure and enable rip routing on R3 follow these commands

```
Router>enable
Router#configu
re terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface
fastethernet 0/0 R3(config-
if)#ip address 30.0.0.2
255.0.0.0 R3(config-if)#no
shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
R3(config-if)#interface serial
0/0 R3(config-if)#ip address
40.0.0.1 255.0.0.0
R3(config-if)#clock rate 64000
R3(config-
if)#bandwidth 64
R3(config-if)#no
shutdown
%LINK-5-CHANGED: Interface Serial0/0, changed state to down
R3(config-if)#exit
%LINK-5-CHANGED: Interface Serial0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed
state to up
R3(config)#router rip
R3(config-router)#network 30.0.0.0
```

```
R3(config-
router)#network
40.0.0.0 R3(config-
router)#exit
R3(config)#
```

To configure and enable rip routing on R4 follow these commands Router>enable

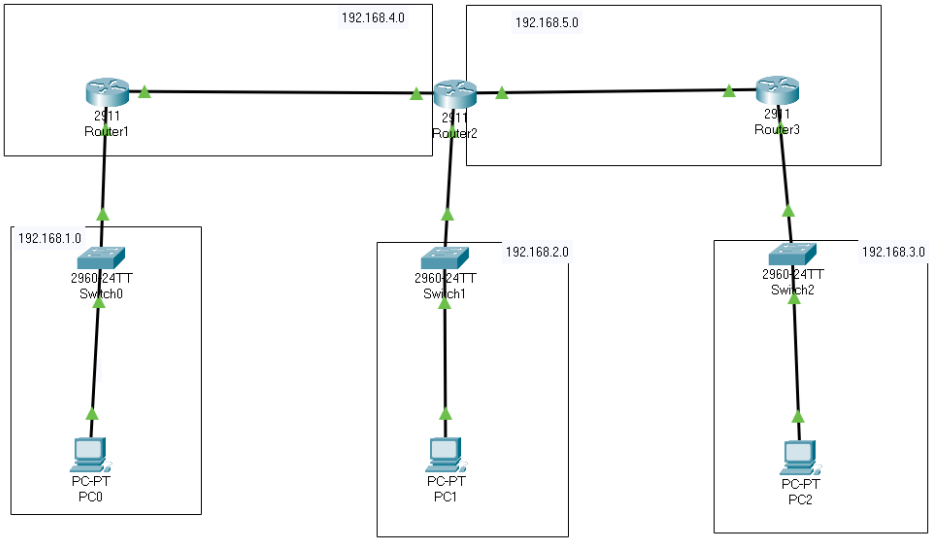
```
Router#configure terminal
Enter configuration commands, one per line.
                                                E
nd with CNTL/Z. Router(config)#interface
serial 0/0/0
Router(config-if)#ip address 40.0.0.2 255.0.0.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up Router(config-if)#exit
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 50.0.0.1
255.0.0.0 Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
Router(config-if)#exit
R4(config)#router rip
R4(config-router)#network 40.0.0.0
R4(config-
router)#network
50.0.0.0 R4(config-
router)#exit
R4(config)#
```

Task4: You can verify that RIP is running successfully via show ip protocols command in privilege mode.

```
R1#show ip
protocols
Routing
Protocol
is "rip"
Sending updates every 30 seconds, next due
in 2 seconds Invalid after 180 seconds, hold
down 180, flushed after 240 Outgoing update
filter list for all interfaces is not set
Incoming update filter list for all
interfaces is not set Redistributing: rip
Default version control: send version 1,
  receive any version
  Interface             Send  Recv  Triggered
  RIP                   Key-chain
```

```
FastEthernet0/0        1      2  1
Serial0/0/0            1      2  1
 Automatic network summarization
 is in effect Maximum path: 4
 Routing for Networks: 10.0.0.0
20.0.0.0
 Passive Interface(s):
 Routing Information Sources:
        Gateway         Distance      Last Update
        20.0.0.2              120      00:00:20
 Distance: (default is 120)
 R1#
```

Configuration need to be done by Students



CONCLUSIONS:

We have successfully configured Router for dynamic routing.